Machine Learning

Support Vector Machine, Part 3

Dong-Kyu Chae

PI of the Data Intelligence Lab @HYU
Department of Computer Science
Hanyang University

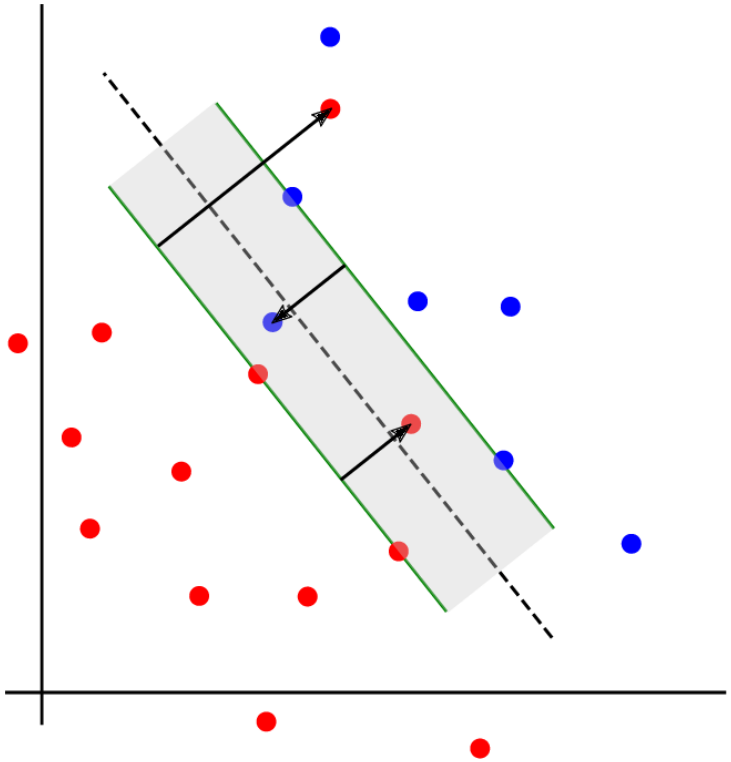
Data
Intelligence
LAB

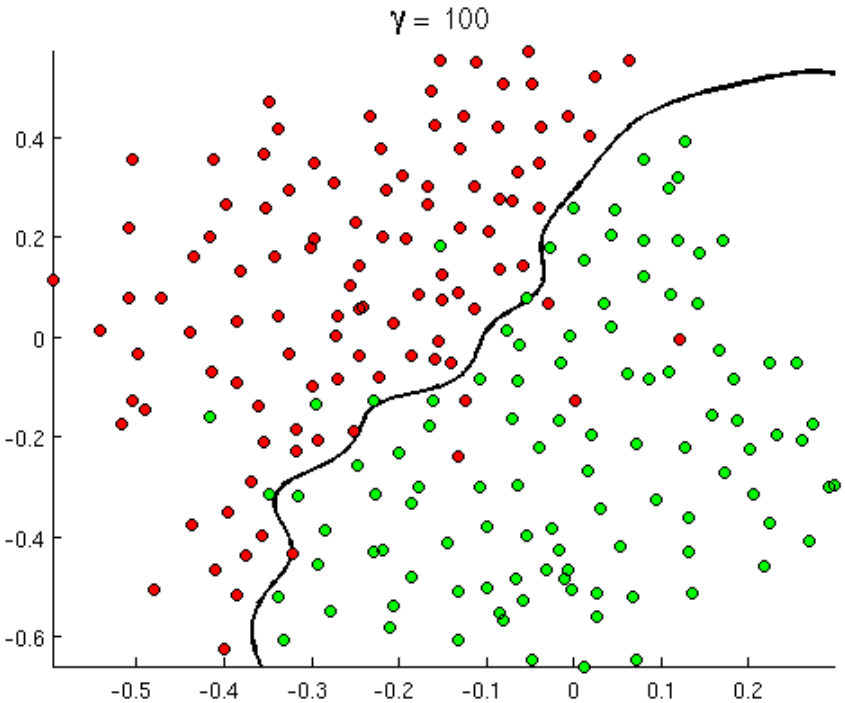
Contents

- **What is SVM ?**
- **Objective function of SVM**
- **Optimization with constraints**
  - Lagrange multiplier
  - Application to SVM
- **Soft margin SVM**
- **Kernel trick**
  - Application to SVM

Non-Separable Cases

- **How to make it work for linearly inseparable cases?**
  - Option 1: Allow the error cases and penalty them
  - Option 2: Non-linear decision boundary



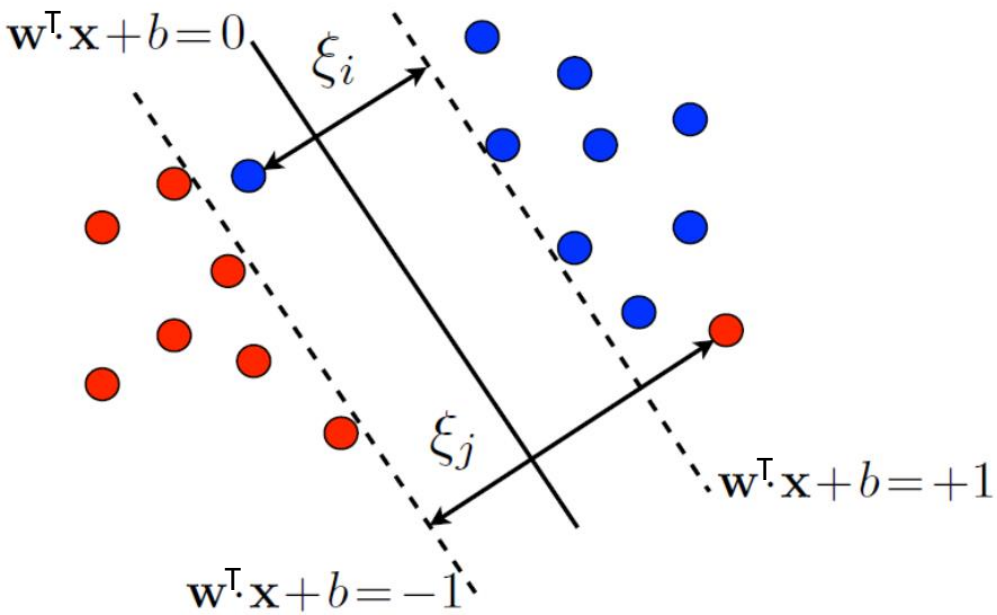


Soft Margin SVM

- **Idea: allow the error cases, just penalty them**

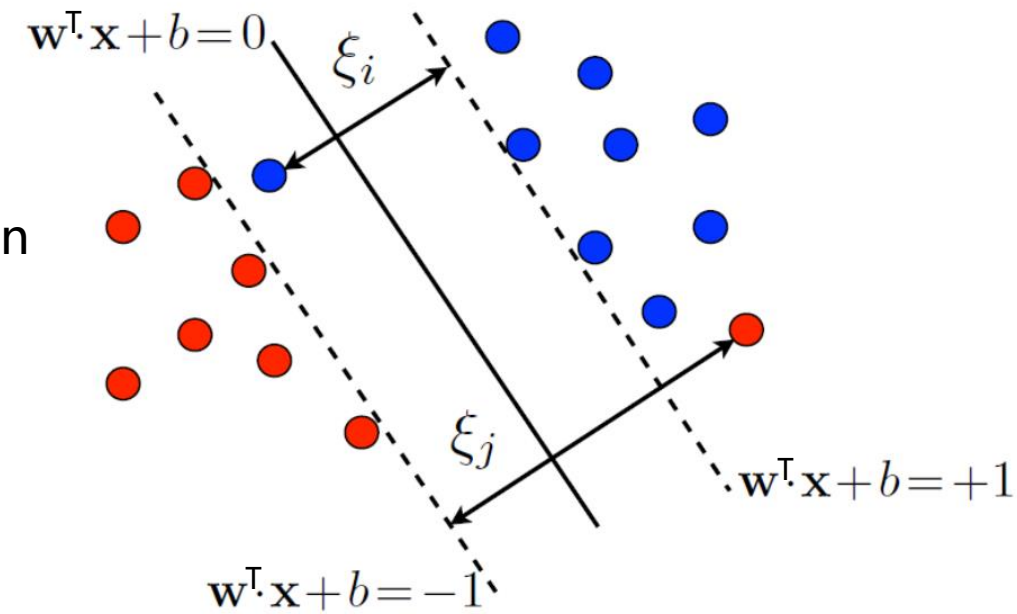
- **Still, there are support vectors and non support vectors**

- **We consider three types of support vectors:**
  1. **On** the margin
  2. **Inside** the margin
  3. **Outside** the margin



Slack Variable

- **How to measure the significance of error?**
  - A set of slack variables $\boldsymbol{\xi} = \{\xi_1, \ldots, \xi_N\}$ is introduced with $\xi_i \geq 0$
    - Each slack variable ξ_i measures the degree of violation of the original constraint
  - Non support vectors
    - $\xi_i = 0$
  - Support vectors on the margin
    - $\xi_i = 0$
  - Support vectors inside and outside the margin
    - $\xi_i > 0$

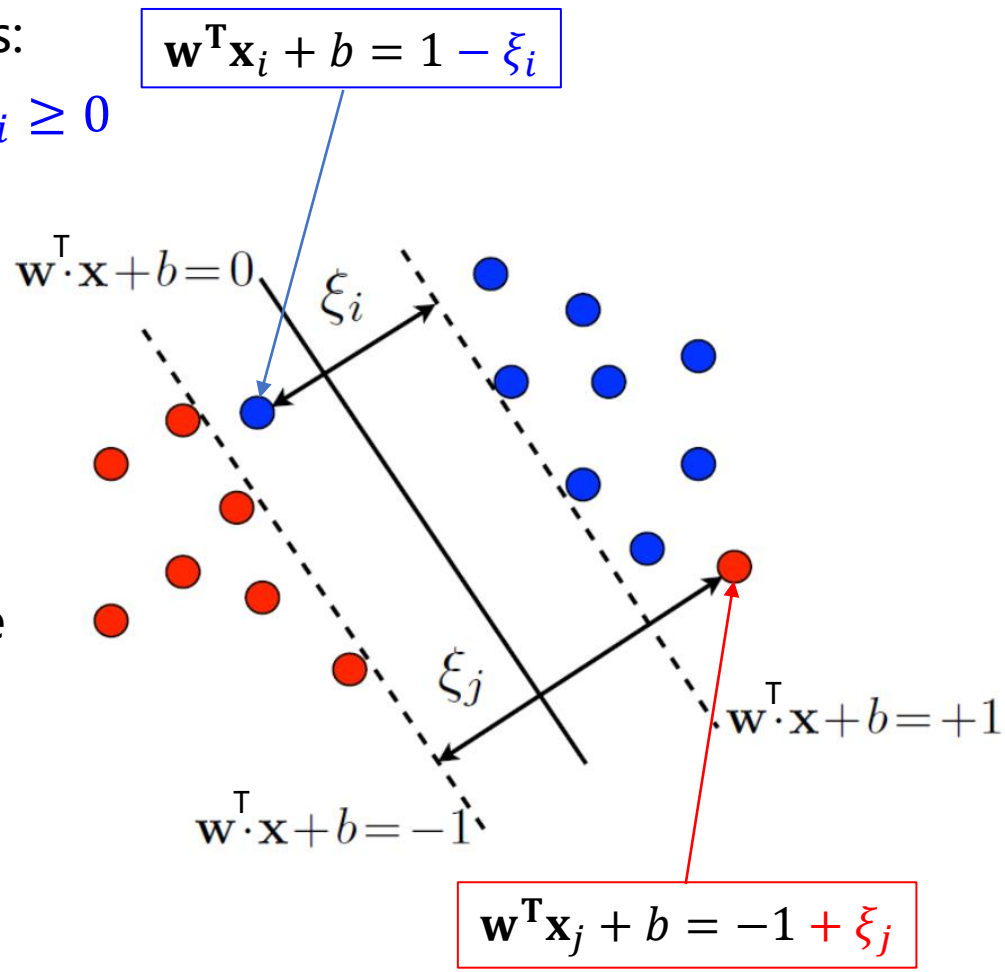


New Objective

- **How to apply those slack variables to the objective function?**
  - New, much relaxed constraints:
    - $y_i(\mathbf{w}^{\mathrm{T}}\mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$
  - New objective:

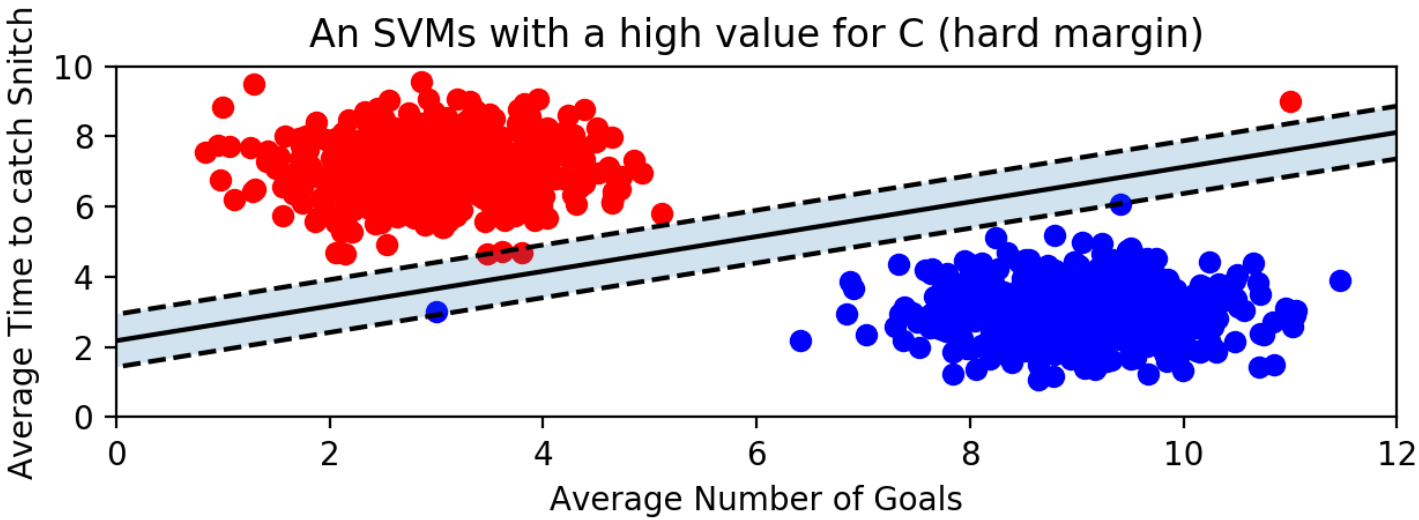
$$\min \frac{||\mathbf{w}||^2}{2} + C\sum_{i=1}^{N} \xi_i$$

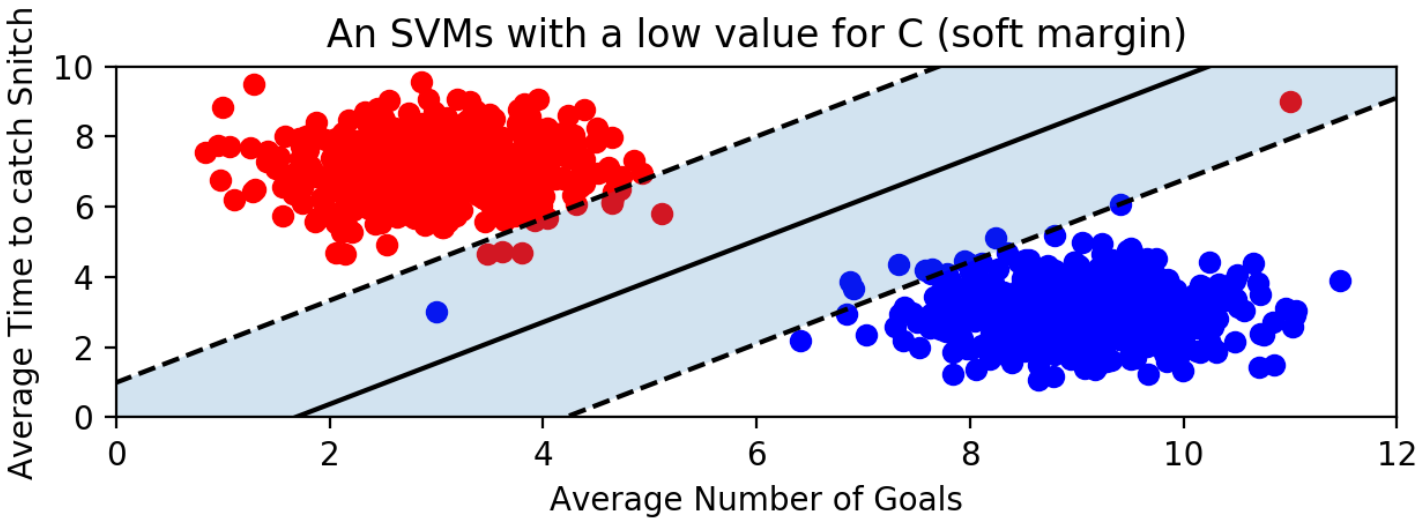
    - C controls the importance of the penalty term
    - C is a hyper-parameter



Results

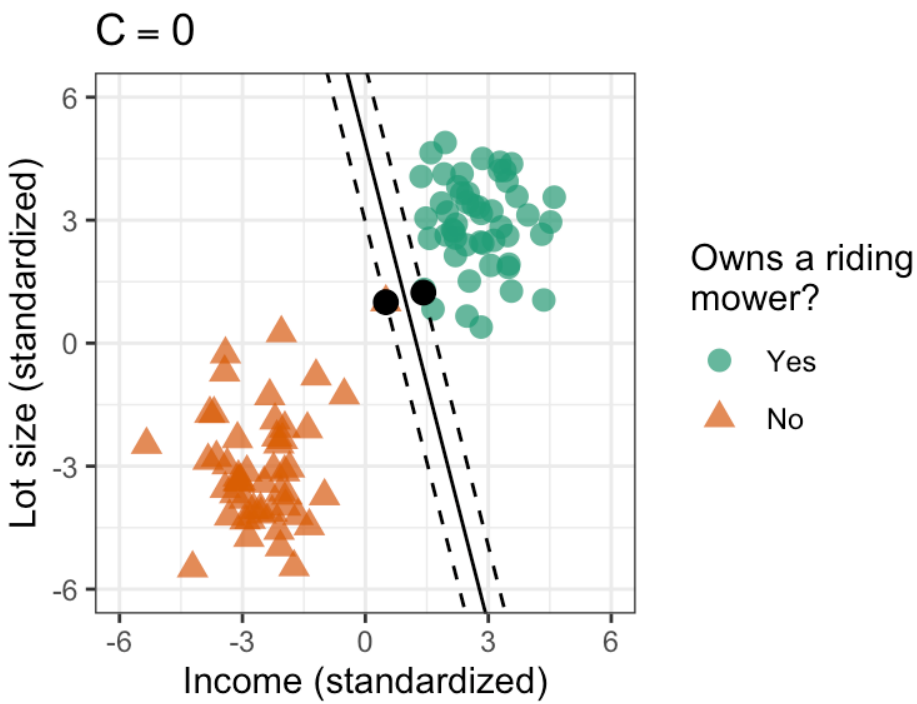
- **Change of decision boundary according to different C**



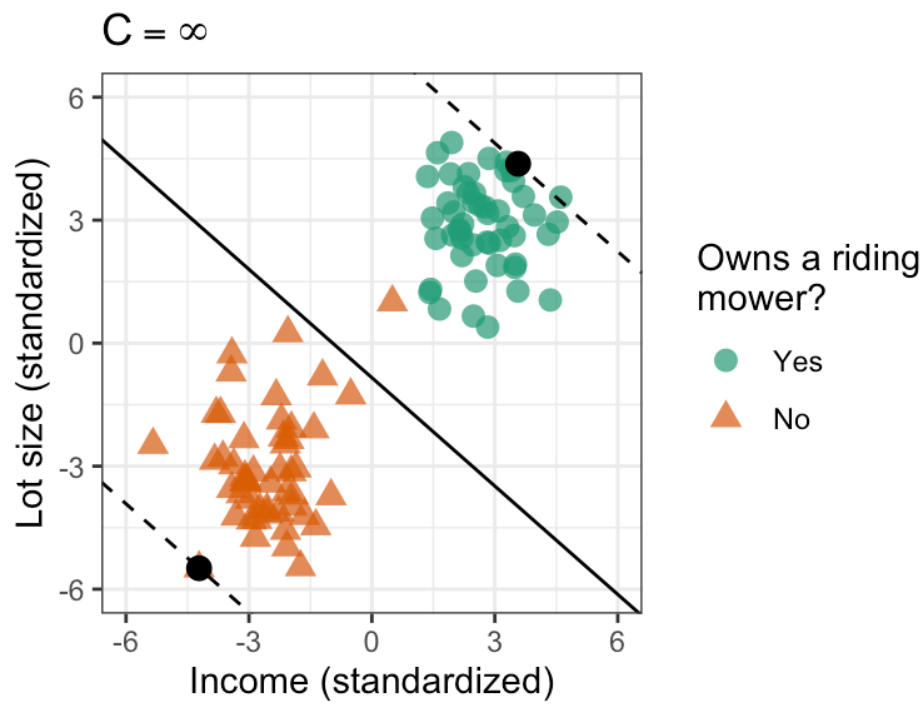


Results

- **Change of decision boundary according to different C (extreme cases)**



(overfitting)



(underfitting)

Optimization

- **New optimization problem:**

$$\textbf{Minimize} \quad \frac{||\mathbf{w}||^2}{2} + C\sum_{i=1}^{N} \xi_i$$

subject to $\xi_i \geq 0$ and $y_i(\mathbf{w}^\intercal \mathbf{x}_i + b) - 1 + \xi_i \geq 0$, $\forall i \in [\mathbf{1}, \boldsymbol{N}]$

- **Largrange primal form:**

$$L(\mathbf{w}, b, \boldsymbol{\alpha}, \boldsymbol{\xi}, \boldsymbol{\gamma}) = \frac{||\mathbf{w}||^2}{2} + C\sum_{i=1}^{N} \xi_i - \sum_{i=1}^{N} \alpha_i\{y_i(\mathbf{w}^\intercal \mathbf{x}_i + b) - 1 + \xi_i\} - \sum_{i=1}^{N} \gamma_i\, \xi_i$$

Optimization

- **Minimizing the primal form**

Lagrangian Primal

$$\max_{\alpha,\gamma} \min_{w,b,\xi} \mathcal{L}(w,b,\alpha,\xi,\gamma) = \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n}\xi_i - \sum_{i=1}^{n}\alpha_i\left(y_i\left(w^T x_i + b\right) - 1 + \xi_i\right) - \sum_{i=1}^{n}\gamma_i\xi_i$$

- **Optimal solutions for each variable:**

① $$\frac{\partial\mathcal{L}(w,b,\xi,\alpha,\gamma)}{\partial w} = 0 \Longrightarrow w = \sum_{i=1}^{n}\alpha_i y_i x_i$$

② $$\frac{\partial\mathcal{L}(w,b,\xi,\alpha,\gamma)}{\partial b} = 0 \Longrightarrow \sum_{i=1}^{n}\alpha_i y_i = 0$$

③ $$\frac{\partial\mathcal{L}(w,b,\xi,\alpha,\gamma)}{\partial \xi_i} = 0 \Longrightarrow C - \alpha_i - \gamma_i = 0, i = 1,2,\ldots\ldots,n$$

Optimization

- **Lagrange dual form:**

$$\mathcal{L}(w,b,\alpha,\xi,\gamma) = \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n}\xi_i - \sum_{i=1}^{n}\alpha_i\left(y_i\left(w^T x_i + b\right) - 1 + \xi_i\right) - \sum_{i=1}^{n}\gamma_i\xi_i$$

(We know)

$$= \frac{1}{2}\|w\|_2^2 - \sum_{i=1}^{n}\alpha_i\left(y_i\left(w^T x_i + b\right) - 1\right) + C\sum_{i=1}^{n}\xi_i - \sum_{i=1}^{n}\alpha_i\xi_i - \sum_{i=1}^{n}\gamma_i\xi_i$$

$$= \sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j x_i^T x_j + C\sum_{i=1}^{n}\xi_i - \sum_{i=1}^{n}(\alpha_i + \gamma_i)\xi_i$$

$$= \sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j x_i^T x_j + C\sum_{i=1}^{n}\xi_i - C\sum_{i=1}^{n}\xi_i$$ (Applying ③)

$$= \sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j x_i^T x_j$$

Optimization

- **Constraints:**

- $\sum_{i=1}^{n} \alpha_i y_i = 0,\ \alpha_i \geq 0, \gamma_i \geq 0, and\ C - \alpha_i - \gamma_i = 0$ for all i: 1~N

- Based on $\alpha_i \geq 0, \gamma_i \geq 0, and\ C - \alpha_i - \gamma_i = 0$, we know the range of alpha:

$$\boxed{0 \leq \alpha_i \leq C}$$

Optimization

- **As a result, the dual form becomes:**

$$\max \quad L_D(\alpha_i) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^{\mathrm{T}} \mathbf{x}_j$$

$$s.t. \quad \sum_{i=1}^{N} \alpha_i y_i = 0 \quad and \quad \boxed{0 \leq \alpha_i \leq C}$$

- Compared to the hard margin SVM, the only difference is the range of α_i

$$\max \quad L_D(\alpha_i) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^{\mathrm{T}} \mathbf{x}_j$$

$$s.t. \sum_{i=1}^{N} \alpha_i y_i = 0 \quad and \quad \alpha_i \geq 0$$

- The quadratic programming is used for getting optimal $\boldsymbol{\alpha} = < a_1, a_2, \ldots, a_N >$

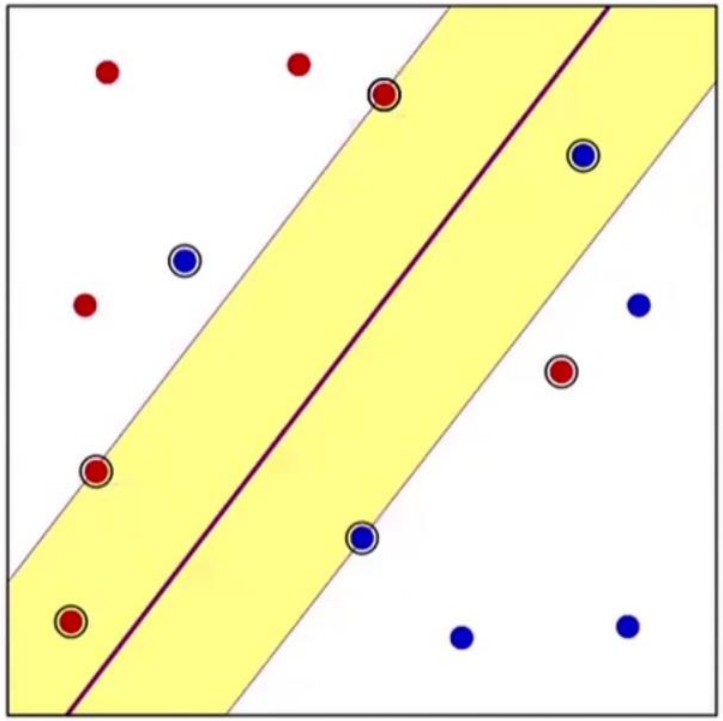
Analysis

- margin support vectors $(0 < \alpha_t < C)$
  - ordinary sv's among C1 points

$$y_t(\mathbf{w}^\top \mathbf{x}_t + b) = 1 \qquad (\xi_t = 0)$$

- non-margin support vectors $(\alpha_t = C)$
  - C2 and C3 points

$$y_t(\mathbf{w}^\top \mathbf{x}_t + b) < 1 \qquad (\xi_t > 0)$$



*note: complementary slackness $\gamma_i \cdot \xi_i = 0$

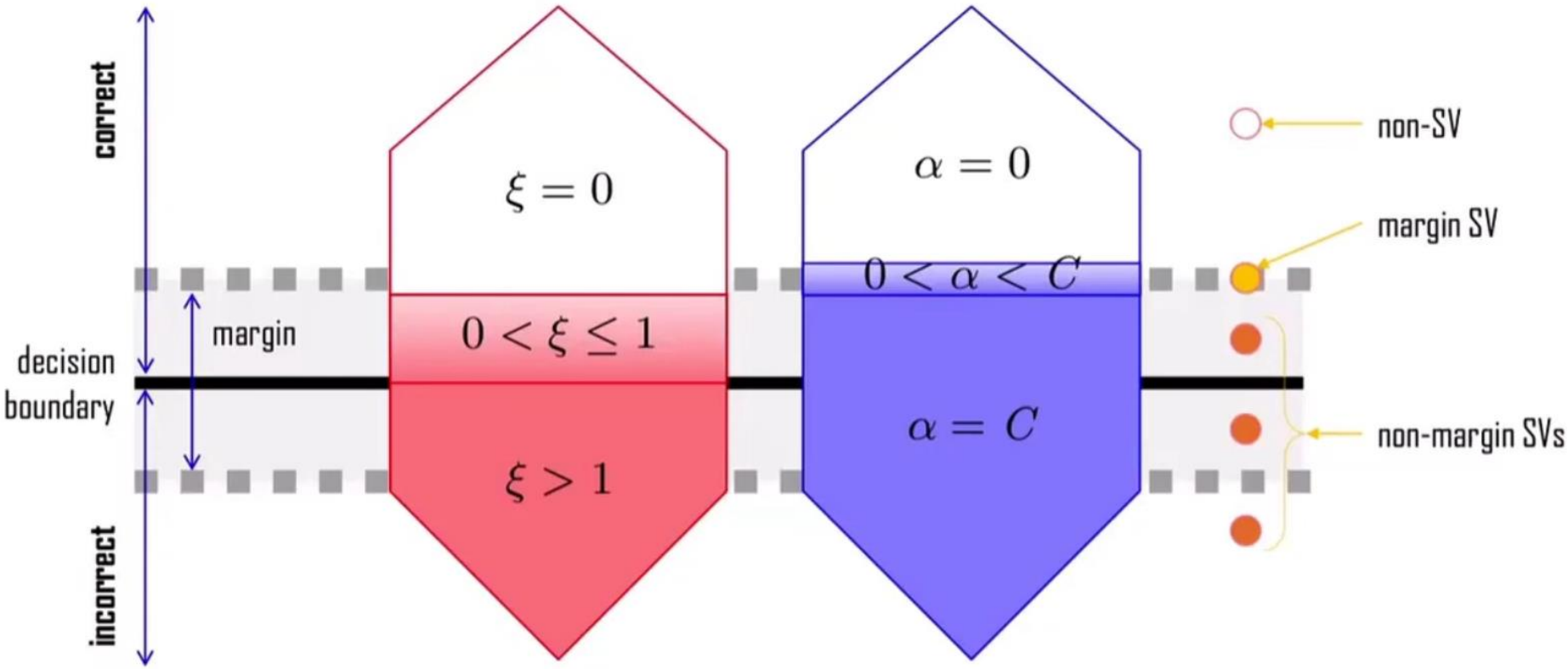
$C = \gamma_i + \alpha_i$

Analysis

*note: $\mathbf{w}^{\mathbf{T}}\mathbf{x}_i + b = 1 - \xi_i$

$0 \leq \alpha_i \leq C$

- **Values of α and ξ for each case**



- $\xi > 0 \iff \alpha = C$
- $\xi = 0 \iff \alpha < C$

Analysis

Example

1. On the margin:

$$C > \alpha_i > 0, \xi_i = 0$$

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) = 1$$

$$\alpha_{11} = 0.44; \xi_{11} = 0$$

$$\alpha_1 = 2.85; \xi_1 = 0$$

2. Inside the margin:

$$\alpha_i = C; 0 < \xi_i < 1$$

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) \le 1$$

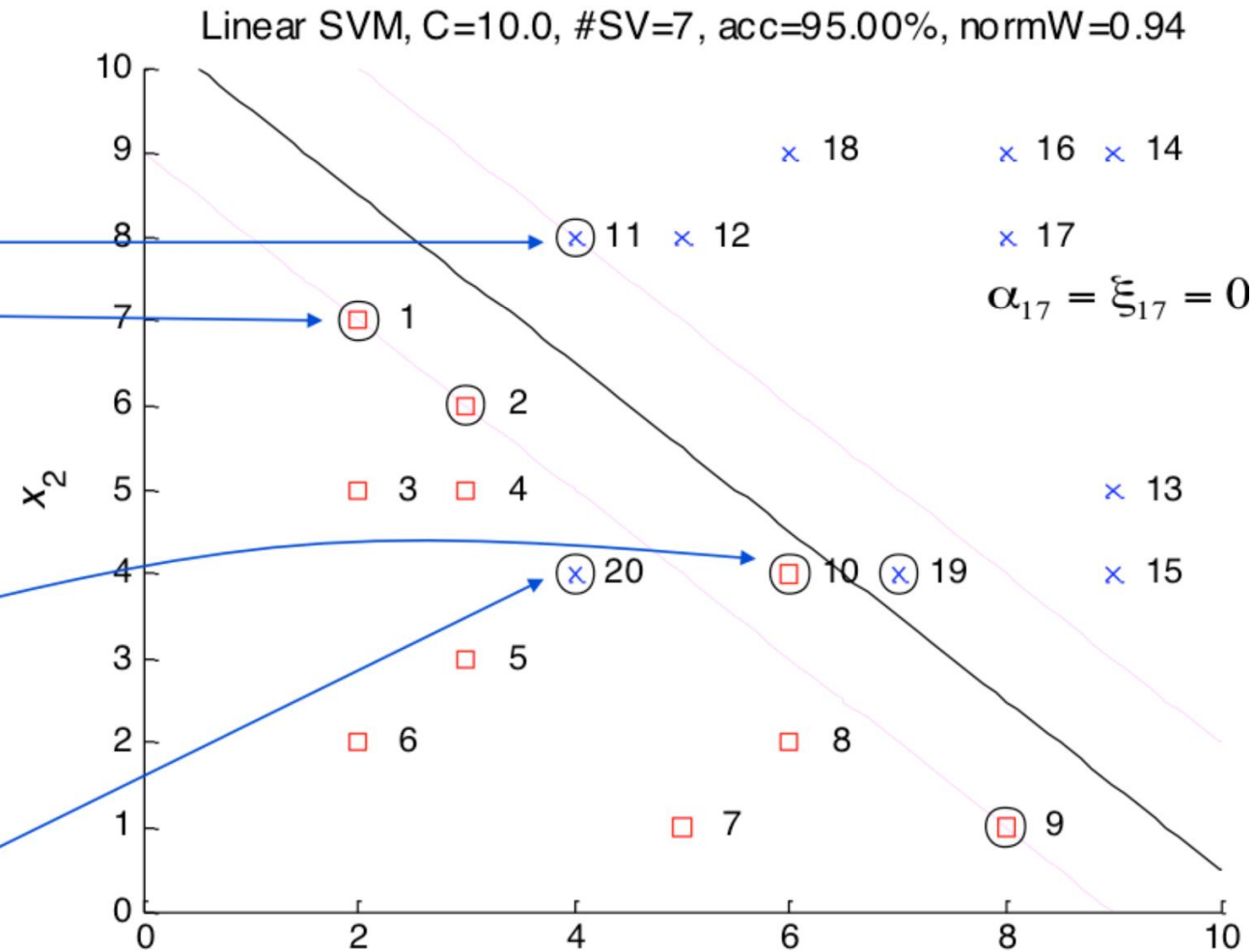
$$\alpha_{10} = 10; \xi_{10} = 0.667$$

3. Outside the margin:

$$\alpha_i = C; \xi_i \ge 1$$

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) \le 1$$

$$\alpha_{20} = 10; \xi_{20} = 2.667$$



Thank You